- Easy to solder
- Cost of the conductor is low and is easily available to buy

2.3 INSULATORS

Insulator is a non-conducting material, i.e. it resists electricity. It has high resistance value normally in mega ohms.

Insulators should be good in condition, then only they prevent electric shock due to current leakage. Generally, insulators are classified into three types. Some examples of the three types of insulators are:

i) **Hard Insulators:** Bakelite, porcelain, wooden plank, glass, mica, etc.
ii) **Soft Insulators:** Rubber, poly-vinyl chloride, varnish coated papers, etc.
iii) **Liquid Insulators:** Mineral oil, shellac, varnish, etc.



Fig 2.1 *Hard Insulator*

Properties of Insulators

The following are the important properties of insulating materials. They are:

- High resistance and specific resistance
- High dielectric strength
- Good mechanical strength
- It withstands high temperature
- May not get change in the shape due to high temperature
- It does not absorb water
- It can be made to any shape
- It does not catch fire easily

2.4 ELECTRICAL TERMS

Here we study the electrical fundamental terms used in electrical engineering.

2.4.1 Current

Flow of electron in a conductor is called as current. It is represented by the letter 'I' and the unit is called ampere(A). Current can be measured by ammeter.

1 Ampere: "One coulomb charge crossing over the area of cross section of the conductor in one second is called 1 ampere.

1 Coulomb: A collection of $2\pi \times 10^{18}$ electrons has a charge of one coulomb.

Example :- For water to flow through a pipe from one end to the required place, some pressure is required. In the same way, electric pressure is required to move the electron from one end to another end. This flow of electron is called current and the pressure required to move the electron is called electric pressure or voltage.

2.4.2 Voltage

The electric pressure which is used to move electrons from one end to another end

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